Module Review: γ-Glutamyl-Putrescine Catabolism (Puu pathway) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Parent bucket:** KEGG ppu00330 "Arginine and proline metabolism" (amino_acid_metabolism) **Module under review:** `gamma_glutamyl_putrescine_catabolism` **Review type:** species-aware module satisfiability + gene-annotation curation

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1. Executive summary

- The γ-glutamyl-putrescine (Puu) catabolic module is **SATISFIABLE (covered)** in *P. putida* KT2440. Every enzymatic step of the canonical *E. coli* six-step Puu pathway (Kurihara et al. 2005, P 15590624) has at least one KT2440 ortholog in the UniProt reference proteome, and most steps are **redundantly encoded** (2–3 paralogs) in two coherent gene clusters.
- **The supplied candidate-gene metadata is incomplete and must be revised.** It captures only PuuB (PP_2448) and the *transaminase-branch* enzyme SpuC-II (PP_5182) for this module, and **omits the two diagnostic committed steps** — the γ-glutamylputrescine synthetase/ligase (PuuA) and the γ-glutamyl-GABA hydrolase (PuuD). A naive satisfiability check on the metadata alone would produce a **false gap**.
- The module boundary should explicitly exclude the **SpuC putrescine:pyruvate transaminase branch** (the "non-glutamylation" route), which is the *major* putrescine route in *Pseudomonas* and shares only the downstream GABA node with the Puu branch.
- **PuuC (step 3)** is resolved by KEGG orthology to **PP_2589** (KO K09472, EC 1.2.1.99) → covered. Only **PuuE (GABA aminotransferase, step 5)** remains **candidate_uncertain**; the previously assumed candidate PP_2588 is actually KEGG-mapped to the transaminase branch (K12256), so the true GABA-AT is not yet pinned.

- **Root cause of the metadata gap identified:** the KEGG ppu bucket assigns no gene to the PuuA (K09470) or PuuD (K09473) orthologs, so KEGG-derived buckets structurally miss the committed steps that UniProt (EC 6.3.1.11 / 3.5.1.94) captures — curate from UniProt EC evidence, not KEGG KO alone.
- Evidence is a mix of **strong homology** + **operon synteny** (direct genome/proteome facts for KT2440) and **functional transfer from *E. coli* K-12 and *P. aeruginosa* PAO1**; no KT2440-specific enzyme assays for the Puu branch were located.

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2. Target-organism pathway definition

Process included: aerobic catabolism of the diamine **putrescine** (1,4-diaminobutane) to **succinate** via *γ-glutamylated intermediates*, feeding carbon/nitrogen into the TCA cycle through the GABA node. The committed, module-diagnostic chemistry is the ATP-dependent **γ-glutamylation of putrescine** and the eventual **hydrolytic removal of the γ-glutamyl group**, which distinguish this route from all other amine catabolism.

Alternate names / database definitions: - "Puu pathway" / "putrescine utilization pathway involving γ-glutamylated intermediates" (*E. coli* nomenclature; Kurihara 2005). - In *Pseudomonas* the orthologous genes carry **spu** ("spermidine/putrescine utilization") names (Lu et al. 2002, P 12081945); KT2440 uses a **mixed puu/spu** annotation. Curators must reconcile the two nomenclatures. - KEGG places these reactions inside map00330 "Arginine and proline metabolism"; there is no standalone KEGG module for the γ-glutamyl branch, which is why a bespoke module document is appropriate.

Neighboring processes to keep SEPARATE from this module: - **SpuC transaminase branch** (putrescine → 4-aminobutyraldehyde → GABA, no γ-glutamylation) — parallel, not part of the γ-glutamyl module. - **GABA shunt** (GABA → succinate semialdehyde → succinate) — shared downstream sink; not diagnostic for putrescine and shared with 5-aminovalerate/other amine catabolism. - **Arginine/agmatine → putrescine** supply reactions (aguA, speA, speB, PP_4523) — upstream of this module (putrescine *biosynthesis*/supply), belong to arginine metabolism proper. - **Arginine succinyltransferase (AST) pathway** (astA/B/D/E, argD), **proline metabolism** (proA/B/I, putA, pip), **ornithine/creatinine/opine** modules — all co-bucketed in ppu00330 but mechanistically unrelated to putrescine γ-glutamylation.

3. Expected step model

Step	Reaction	Enzyme (EC)	KT2440 gene(s)	Status
0	Putrescine import	PuuP permease; Pot/Spu ABC	**puuP PP_1229**; potA PP_0411, spuF/ potA PP_5179	covered
1	Putrescine + Glu + ATP → γ-glutamylputrescine	**PuuA / glutamate-putrescine ligase, EC 6.3.1.11; KEGG K09470**	**PP_2178 (puuA-I), PP_5299 (puuA-II)**; spuB PP_5183, spuI PP_5184 (glutamylpolyamine synthetases)	covered — **missing from metadata & from KEGG ppu K09470**
2	γ-glutamylputrescine + O_2 → γ-glutamyl-γ-aminobutyraldehyde	**PuuB / γ-glutamylputrescine oxidase** (EC 1.4.3.–; KEGG K09471)	**PP_2448 (puuB)** ✓ + KEGG paralogs PP_3146, PP_4548, PP_5273	covered
3	→ γ-glutamyl-GABA ($NAD(P)^+$)	**PuuC / 4-(γ-glutamylamino)butanal dehydrogenase, EC 1.2.1.99; KEGG K09472**	**PP_2589 (unique KEGG K09472 assignment)**	**covered** (upgraded from uncertain)
4	γ-glutamyl-GABA + H_2O → GABA + Glu	**PuuD / γ-glutamyl-γ-aminobutyrate hydrolase, EC 3.5.1.94; KEGG K09473**	**PP_5298 (puuD), PP_2179 (spuA), PP_3598**	covered — **missing from metadata & from KEGG ppu K09473**
5	GABA + 2-oxoglutarate → succinate semialdehyde + Glu	**PuuE / GabT, EC 2.6.1.19**	true GABA-AT unassigned; PP_2588 is KEGG-mapped to the *transaminase branch* (K12256), not this step	**candidate_uncertain**
6	Succinate semialdehyde → succinate (shared GABA shunt)	SSADH, EC 1.2.1.24/1.2.1.79	sad-I PP_2488, sad-II PP_3151, gabD-II PP_4422	covered (shared)
Reg		PuuR	**puuR PP_5268**	present

Step	Reaction	Enzyme (EC)	KT2440 gene(s)	Status
	Putrescine-responsive repression			

Gene-cluster architecture (direct KT2440 genome evidence): - Cluster A (PP_2178–2180): *puuA-I* (PP_2178) – *spuA/puuD* (PP_2179) – *spuC-I* (PP_2180). Co-locates a γ-glutamyl synthetase, a γ-glutamyl-GABA hydrolase, and a polyamine transaminase. - **Cluster B (PP_5182–5184 + PP_5298–5299, divergent):** *spuC-II* (PP_5182) – *spuB* (PP_5183) – *spuI* (PP_5184) … *puuD* (PP_5298) – *puuA-II* (PP_5299), with regulator *puuR* (PP_5268) nearby. This mirrors the divergent *spuABCDEFGH–spuI* locus of *P. aeruginosa* PAO1 (Lu 2002, P 12081945).
- **PuuB (PP_2448) is standalone** (not syntenic with A or B), and the GABA-AT/ALDH candidates PP_2588–2589 sit in a separate transporter-linked cluster (PP_2585–2589).

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4. Candidate genes and evidence

High-confidence, module-core (should be marked covered): - PP_2448 `puuB` — γ-glutamylputrescine oxidase. UniProt Q88K44 name matches the diagnostic step 2 enzyme. *In the candidate list.* Evidence: strong homology to characterized *E. coli* PuuB; the reaction is unique to this module → high curation value. **Promote to fetch-gene review** as the module anchor. - **PP_2178 `puuA-I` / PP_5299 `puuA-II` — glutamate-putrescine ligase (EC 6.3.1.11).** UniProt-annotated with the exact EC of *E. coli* PuuA (P78061). Committed first step; two paralogs. **Not in candidate metadata — add and promote. - PP_5298 `puuD` / PP_2179 `spuA` / PP_3598 — γ-glutamyl-γ-aminobutyrate hydrolase (EC 3.5.1.94).** All three carry the curated UniProt FUNCTION "Involved in the breakdown of putrescine via hydrolysis of the gamma-glutamyl linkage of gamma-glutamyl-gamma-aminobutyrate." **Not in candidate metadata — add and promote. - spuB PP_5183 / spuI PP_5184 — glutamylpolyamine synthetases.** PuuA-family; provide redundancy/broadened substrate range (spermidine as well as putrescine). Add as supporting evidence for step 1.

Present but branch-adjacent (curation caveat): - **PP_5182 `spuC-II` (in candidate list) and PP_2180 `spuC-I` — polyamine:pyruvate transaminase.** These are the **non-glutamylation** route enzymes (SpuC = putrescine:pyruvate aminotransferase, biochemically demonstrated in

PAO1, Lu 2002). They belong to the parallel transaminase branch; **do not count SpuC as a γ-glutamyl-module step.** Their presence in the same operons explains why the two routes are annotation-entangled in *Pseudomonas*.

Ambiguous / broad-annotation candidates (candidate_uncertain): - **PuuC step (ALDH):** PP_2589 (generic "aldehyde dehydrogenase family"), prr PP_2801 (EC 1.2.1.19), patD PP_1481 (EC 1.2.1.19/1.2.1.54), kauB PP_5278 (EC 1.2.1.54). *E. coli* PuuC (P23883) is a broad-specificity ALDH, so cross-mapping is soft. No unique assignment; likely functional redundancy. - **PuuE/GABA-AT step:** PP_2588 "aminotransferase, class III" (no EC in UniProt) is the leading candidate by family and by operon context (adjacent to ALDH PP_2589 and transporter PP_2586). *aruH* PP_3721 and other class-III AATs could cross-react. GABA transamination is shared with the GABA shunt, so this step is not diagnostic.

Genes in the candidate list that do NOT belong to this module (bucket noise from ppu00330): aguA, speA, speB, PP_4523 (agmatine→putrescine *supply*); speC, ldcC, nspC (other decarboxylases); aruH/aruI, ooxA/ooxB, creA, codA (arginine/opine/creatinine); astA-E, argD (AST pathway); proA/B/I, putA, pip, oplA/B, ocd (proline/5-oxoproline/ornithine); PP_4983 (Trp); PP_3146/PP_5273/PP_4548 generic oxidoreductases; PP_2932 amidase (ppu00643). These are correctly outside the γ-glutamyl-putrescine module.

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5. Gaps, ambiguities, and likely over-annotations

- **Root cause of the metadata gap (KEGG under-annotation):** KEGG REST `link/ppu` returns **no gene** for KO **K09470 (PuuA, EC 6.3.1.11)** or **K09473 (PuuD, EC 3.5.1.94)**, even though UniProt annotates PP_2178/PP_5299 (PuuA) and PP_5298/PP_2179/PP_3598 (PuuD) with those exact ECs and curated FUNCTION statements. Because the candidate bucket is KEGG-derived, it inherited these omissions. **Recommendation: build/patch the bucket from UniProt EC + curated FUNCTION, not KEGG KO alone.**
- **Metadata gap (not a biological gap):** committed steps 1 (PuuA) and 4 (PuuD) are absent from the candidate list but **present in the genome** → classify as `module_needs_revision`, not `gap`.
- **PuuC step now RESOLVED:** KEGG KO **K09472** (4-(γ-glutamylamino)butanal dehydrogenase, EC 1.2.1.99) maps **uniquely to PP_2589** → mark step 3 **covered** (no longer candidate_uncertain). prr/patD/kauB are not the PuuC ortholog.

- **PP_2588 correction:** KEGG maps PP_2588 to **K12256 (putrescine:pyruvate transaminase, EC 2.6.1.113)** together with PP_5182 — i.e., the **transaminase branch**, not PuuE. The genuine PuuE/GABA-AT (EC 2.6.1.19) remains unassigned → step 5 stays `candidate_uncertain`.
- **PuuB paralog spread:** KEGG K09471 maps PP_2448 **plus** the generic "oxidoreductase" candidates PP_3146, PP_4548, PP_5273; these are over-broad KO propagations that should be gene-reviewed rather than auto-counted.
- **Paralog ambiguity:** steps 3 (PuuC ALDH) and 5 (PuuE GABA-AT) → `candidate_uncertain`. Redundancy is real (multiple ALDHs/AATs), so flux-carrying paralog cannot be assigned from sequence alone.
- **Branch mis-assignment / boundary error:** `spuC-II` (PP_5182) is bucketed as a core step but is the transaminase-branch enzyme → boundary clarification needed.
- **Likely over-propagated / broad EC-GO mappings:** kauB (EC 1.2.1.54, primarily guanidinobutyraldehyde/arginine catabolism), patD, prr are pulled in by generic ALDH EC terms; PP_2588 lacks any EC. These should not be auto-counted as module evidence without gene-level review.
- **Shared downstream node:** SSADH (PP_2488/PP_3151/PP_4422) and GABA-AT are shared with 4-aminobutyrate/5-aminovalerate catabolism; counting them as module-specific inflates coverage.

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6. Module and GO-curation recommendations

Per-step calls: - Step 0 import — **covered** (puuP PP_1229; Pot/Spu ABC). - Step 1 γ-glutamylation (PuuA) — **covered**; **revise metadata** to add PP_2178, PP_5299, PP_5183, PP_5184. - Step 2 oxidase (PuuB) — **covered** (PP_2448). - Step 3 ALDH (PuuC) — **covered** (PP_2589; unique KEGG K09472 / EC 1.2.1.99). - Step 4 hydrolase (PuuD) — **covered**; **revise metadata** to add PP_5298, PP_2179, PP_3598 (KEGG K09473 unassigned in ppu). - Step 5 GABA-AT (PuuE) — **candidate_uncertain**; PP_2588 reassigned to transaminase branch (KEGG K12256), true GABA-AT still to be identified. - Step 6 SSADH — **covered (shared)**; annotate as non-diagnostic.

Module-level actions: 1. Set module status = **covered / satisfiable** with an operon-supported note, but flag the **metadata as `module_needs_revision`** because the committed enzymes are missing from the candidate list. 2. **Create/curate a dedicated module document** for γ-glutamyl-putrescine catabolism (KEGG has no standalone module), with explicit boundaries: include steps 1–5 (γ-glutamyl branch), reference-only for import and SSADH, and **exclude the SpuC transaminase branch** (link it as a sibling module "putrescine transaminase catabolism"). 3. **GO term usage:** apply GO:0009447 (putrescine catabolic process) and, for step 1/4, the specific activities glutamate–putrescine ligase (GO:0034024 / EC 6.3.1.11) and γ-glutamyl-γ-aminobutyrate hydrolase (EC 3.5.1.94). Request/confirm a GO term for γ-glutamylputrescine oxidase (PuuB) activity if absent. Avoid attaching broad ALDH/aminotransferase GO terms as module evidence. 4. Record the **puu/spu nomenclature reconciliation** so PP_2179 (`spuA`) = PuuD, PP_5183/5184 (`spuB/spuI`) = PuuA-family, PP_2180/5182 (`spuC`) = transaminase branch.

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7. Genes to promote to full `fetch-gene` review

Priority order: 1. **PP_2448** **`puuB`** — module anchor (diagnostic step 2). *Already in list.* 2. **PP_2178** **`puuA-I`** and **PP_5299** **`puuA-II`** — committed step 1; **add to module**. 3. **PP_5298** **`puuD`** (and **PP_2179** **`spuA`**, **PP_3598**) — committed step 4 hydrolase; **add to module**. 4. **PP_2588** (class III aminotransferase) — resolve PuuE/GABA-AT assignment; needs EC and substrate evidence. 5. **PP_2589** (aldehyde dehydrogenase family) — resolve PuuC assignment vs prr/patD/kauB. 6. **PP_5182** **`spuC-II`** (and **PP_2180** **`spuC-I`**) — reassign from core step to the transaminase sibling branch. 7. **spuB PP_5183** / **spuI PP_5184** — confirm glutamylpolyamine-synthetase role and substrate range.

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8. Key references

- Kurihara S. et al. (2005) *A novel putrescine utilization pathway involves γ-glutamylated intermediates of Escherichia coli K-12.* J Biol Chem.

  P 15590624

  — Defines the canonical six-step Puu pathway (expected step model).
- Kurihara S. et al. (2008) *γ-Glutamylputrescine synthetase in the putrescine utilization pathway of E. coli K-12.*

  P 18495664

— Characterizes PuuA (EC 6.3.1.11), the committed step; shows the Puu route dominates putrescine utilization in *E. coli.*

- Kurihara S. et al. (2009) *The putrescine importer PuuP of E. coli K-12.*

P 19181795

— PuuP importer (ortholog PP_1229).

- Lu C-D. et al. (2002) *Functional analysis and regulation of the divergent spuABCDEFGH-spuI operons for polyamine uptake and utilization in Pseudomonas aeruginosa PAO1.* J Bacteriol.

P 12081945

— Establishes the *Pseudomonas* spu operon architecture and SpuC = putrescine:pyruvate transaminase (branch assignment; strong transfer to KT2440 by synteny).

- Chou H.T. et al. (2008) *Transcriptome analysis of agmatine and putrescine catabolism in Pseudomonas aeruginosa PAO1.* J Bacteriol.

P 18192388

— Shows the SpuC/transaminase route is the major putrescine route and that γ-glutamylation is an alternative "present in some organisms" (boundary rationale).

Evidence provenance: KT2440 gene identities, EC numbers, curated FUNCTION statements, and locus-tag synteny are **direct facts** from the UniProt reference proteome (UP000000556, taxon 160488). Step-to-enzyme functional assignments are **transferred by homology/synteny** from *E. coli* K-12 (Puu) and *P. aeruginosa* PAO1 (Spu); no KT2440-specific enzyme kinetics for the γ-glutamyl branch were found, so all "covered" calls for steps 1–5 rest on homology + operon context, which is **strong** for PuuA/PuuB/PuuD and **uncertain** for the PuuC and PuuE paralog choices.